Merchmix Data Scientist Technical Assignment

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Repository: https://github.com/upashanadutta-maker/merchmix-tech-assignment

I treated this assignment as two linked tasks. First, I forecast four weeks of sales for recently active style-colour groups. Second, I used the top three source styles as references for a constrained agent-based design workflow. I kept the claims separate: model performance applies to existing products, not to the three new concepts.

Final concept board: blue ribbed sweater, black wide-leg trousers, and pink short-sleeve knit top.

1. Forecasting approach

I defined a style as the combination of product code and colour group. The same garment can behave differently across colours, so I kept colour in the modelling unit. The final candidate set includes styles with at least one sale in the previous eight completed weeks.

The target is units sold over the next four weeks, covering 21 September to 18 October 2020. I built 34 features: 25 numerical and 9 categorical. These include recent sales, customer counts, momentum, lifecycle, price and channel summaries, calendar context, and catalogue attributes.

Partition	Snapshot coverage	Rows
Training	18 snapshots, ending 2 March 2020	606,999
Validation	27 April and 25 May 2020	58,265
Held-out test	20 July and 17 August 2020	65,935

I used chronological splits rather than a random split. CatBoost, XGBoost and a recent-four-week baseline were evaluated on the same validation rows. A separate 600-case ETS benchmark was also run, but its absolute errors are not directly comparable with the full validation population.

2. Model results

Validation

Model	MAE	RMSE	Top-20 overlap
CatBoost	23.5288	79.4335	47.5%
XGBoost	23.9814	83.1169	32.5%
Recent-four-week baseline	27.5441	87.6625	40.0%

I selected CatBoost using validation MAE. The held-out test period was not used to select the model or tune the final settings.

Held-out test

Model	MAE	RMSE	Top-20 overlap
CatBoost	15.2722	62.8647	45.0%
Recent-four-week baseline	24.6964	80.8710	35.0%

CatBoost reduced held-out test MAE by 38.2% compared with the baseline. The exported model was then refit after model selection using all eligible labelled history available at the final forecasting cutoff.

3. Selecting the three source styles

The final model scored 32,553 eligible style-colour groups. I used a combined ranking rather than predicted sales alone: 40% forecast-sales percentile, 30% adjusted historical-growth percentile, and 30% recent-customer percentile. The growth component is shrunk when the prior volume is small. This is a practical business rule, not a probability of success.

Rank	Article	Product / source colour	Recent 4w	Forecast 4w
1	0673677002	Henry polo. (1) / Black	1,632	1,667.18
2	0865799006	Pink HW barrel / Light Beige	1,323	1,664.99
3	0915529001	Liliana / Beige	2,227	2,016.44

The third style remains highly ranked even though its forecast is below its recent four-week sales because the final ranking also reflects adjusted momentum and customer breadth.

4. From the source styles to the final concepts

Concept 1: blue ribbed crew-neck sweater

Source: article 0673677002, Henry polo. (1), Black

Kept: regular length, straight hem, simple sweater silhouette.
Changed: black to blue, high/polo neckline to crew neck, plain main body to ribbed surface.

Concept 2: black wide-leg trousers with patch pockets

Source: article 0865799006, Pink HW barrel, Light Beige

Kept: high-waist and five-pocket trouser identity.
Changed: light beige to black, tapered leg to wide leg, added front patch pockets.

Concept 3: pink boat-neck short-sleeve knit top

Source: article 0915529001, Liliana, Beige

Kept: regular length, straight hem, plain knit body.
Changed: beige to pink, square neckline to boat neck, long sleeves to short sleeves.

5. Agent-based workflow

I used a manager agent to coordinate separate stages instead of asking one model to do everything in one prompt. The workflow was:

1. Retrieve forecasting evidence through MCP-compatible tools.
2. Inspect the three source images and save structured observations.
3. Propose exactly two visible design changes per garment using a reusable design skill.
4. Pause for approval of the observations, colours, design changes, and compiled image prompt.
5. Generate one board containing all three concepts.
6. Run an independent visual review without providing the proposed values to the reviewer.
7. Compare the review with the approved plan in Python and save the result.
8. Record final human approval.

The repository preserves the raw agent responses, MCP calls, approvals, prompt, image receipt, hashes, and verification files. The current offline verification run contains 38 passing tests.

6. Limitations and what I would do next

- The H&M data do not include reliable stock-on-hand history, so the model forecasts observed sales rather than unconstrained demand.
- Validation and test each cover two snapshot dates, which limits the evidence across seasons and market conditions.
- The 40/30/30 winner weights are business choices and have not been prospectively validated.
- Historical sales do not prove that a garment feature caused the performance.
- The generated concepts have not been tested for demand, fit, manufacturing feasibility, or cost.
- The visual review flagged two descriptor mismatches: the sleeve label for the blue sweater and the intended ankle length of the black trousers. I kept those records visible rather than hiding them.

With more time, I would add inventory or availability data, run more rolling-origin backtests, and test the concepts with a merchandiser or customers before treating them as product recommendations.

7. How to review the submission

Item	Location
Executed forecasting notebook	merchmixforecasting.ipynb
Agent workflow	merchmix/workflow_v2.py
MCP tools	merchmix/mcp_server.py and merchmix/mcp_client.py
Reusable skill	skills/evidence-led-fashion-concepts/SKILL.md
Saved run evidence	run_002_source_review/
Verification output	verification/pytest_current.txt

Install the two dependency files and run `pytest -q` from the repository root. The raw H&M files are not included; the notebook is configured for Kaggle with the H&M Personalized Fashion Recommendations dataset attached.